Problem Set 3 for AERM Macro

Preliminaries

If you submit in time, I will look at your work and provide some feedback. Please submit via canvas. Your submission will consist of two pieces. One piece is the code that you use to compute your results. This could be several files depending on how you organize your work. The other piece is a cleanly typed (use latex or lyx) pdf that contains all results, derivations, figures, and explanations.

Exercise 1: Income Risk

The goal of this exercise is to enable you to consider realistic income processes in your model. We still want to use value function iteration, so we still have to discretize the income process. We will use the Rouwenhorst method to do this.

Consider the continuous AR(1) process for the log of income

$$y_{t+1} = \rho y_t + u_{t+1}$$

where $\rho \in [0, 1)$.

1. Build a function that computes a stochastic matrix using the Rouwenhorst method, look at the lecture notes to recall how to build this stochastic matrix recursively. The function should take as inputs: persistence ρ, number of discrete states N_y, and the variance of the shock σ_u^2. Note that the distance form highest to lowest log income level is endogenous to these inputs.[1]

2. Use the code to construct a stochastic matrix with parameters $\rho = .9$, $\sigma_u^2 = .015$, and $N_y = 7$. Compute the stationary distribution. Plot the stationary distribution over the **level** of income.

3. Create a sample of 2000 households and simulate their income for a ten year period (initial draw plus 9 more years). Assume that the initial distribution is drawn from the stationary distribution. To simulate the draws, set a seed == 1776, and create the matrix of uniform draws rand_mat_N_T = rand(2000,10). This way, we will all use the same random sequence and we can compare our results exactly.

4. Run a pooled OLS regression of current log income on lagged income and report the coefficient, i.e.,

$$y_{i,t} = \beta y_{i,t-1} + \epsilon_{i,t}.$$

[1] The log income grid space is centered around zero, but that need not worry you since you can pretty much shift actual income in your model by adding a constant to the log, or raising the wage level.

Ideally, $\hat{\beta}$ is close to ρ.

5. Compute the standard errors assuming the errors are iid normal with constant variance, i.e., I want the simple variance $Var\left(\hat{\beta}\right) = (X'X)^{-1}\frac{u'u}{N}$. Is the coefficient you estimated within the 95% confidence interval of the true value $\rho = .9$?

Exercise 2: Income Shocks and Consumption Response

The goal of this exercise is to better understand the findings of the Blundell/Pistaferri/Preston (2008) paper that we discussed in class. To this end, we will simulate a model of optimal consumption in the presence of income risk.

As we do all this, make sure to not get confused between logs and levels. Some steps happen in logs and others in levels.

1. Solve an incomplete market model with income risk. Suppose there are 5 different income states, and use $\rho = .98$ and $\sigma_u^2 = .015$. Further assume the flow utility $u(c) = \frac{c^{-1}}{-1}$, i.e., CRRA with coefficient of relative risk aversion 2, and discount parameter $\beta = .95$.
   We hold the interest rate fixed at 3%, i.e., this is a partial equilibrium model. Further, we assume the wage is one, which is without loss of generality. Use a log linear spaced asset grid B_grid = exp.((LinRange(0,Log(B_upper+1),N)) .-1 where B_upper should be at least 15 times the highest income. Solve for the policy function. Check that the consumption policy function looks reasonable, which means some curvature for low income households, and rich households have a sufficiently large upper bound so they can save if they want to.

2. The policy function is a discrete function of $c(B, y)$. We now use a simple linear interpolation to make the policy function continuous in B. We do this because we don't want weird jumps in consumption and assets because of our discretization. A linear interpolation means that whenever the asset position falls between two grid points, say $B_1 < B < B_2$, the consumption policy is a simple weighted average

$$c(B, y) = c(B_1, y) + \frac{B - B_1}{B_2 - B_1}\left(c(B_2, y) - c(B_1, y)\right).$$

   Code this up as a function in Julia which takes as input the discrete policy function, and some particular value B.

3. Create a new random matrix rand_mat_N_T_2 = rand(2000,20). Suppose for simplicity that all households start out with an asset position of 1.2 and a log income of zero. Simulate the trajectory of household income, and household consumption. The income comes from applying the stochastic income matrix. Consumption and next period's assets follow from the policy

function (which is now continuous thanks to your interpolation). Note that assets need no longer be on the grid and follow from the budget constraint.

4. Add classical measurement error to log income. Suppose the variances of the measurement error is $\sigma^2_{y,error} = .05$ and mean zero. To do this, draw a new random matrix using rand_mat_error = mean .+ std_dev .* randn(2000, 20) where the mean is zero and the standard deviation is the square root of the variance (because the randn command creates a standard normal distribution so we have to rescale it). Now add this matrix to the simulated log income, i.e., we got some true income y and we got income measured with error $\tilde{y} = y + \epsilon$.[2]

5. Discard the first ten years of simulated data. This is helpful because initially all households start out at the same spot in our simulation.[3] But ten years in, this is no longer true and we have a nice cross-sectional distribution (people often drop some early years when they simulate the data).

6. Follow BPP's simple case and compute the moments

$$\phi = \frac{Cov(\Delta c, \Delta y)}{Cov(\Delta y, \Delta y_{-1} + \Delta y + \Delta y_{+1})}$$
$$\psi = \frac{Cov(\Delta c, \Delta y_{+1})}{Cov(\Delta y, \Delta y_{+1})}$$

   Report the parameters you find in your simulated model. Are they at all close to what BPP find in their baseline specification? You should also find that the passthrough from transitory shocks ψ is zero. Why is that?

7. Next we explore how important borrowing constraints are for estimating the "partial insurance" parameters from BPP. Categorize households as borrowing constrained if their asset wealth is less than 5% of their current income, i.e., households with little savings. Recompute the partial insurance parameters for constrained and unconstrained households. What do you find?

8. Lastly, keep only the 5% of households whose average wealth over the entire ten years is the highest. Rerun the regressions, what do you find. What do you think is going on?

[2] We do this because this keeps the analysis a lot easier. If you wanted to replicate the BPP paper exactly you would want to introduce transitory shocks to the actual state space, i.e., you'd need another state variable that represents transitory shocks. Our approach is easier because we treat the transitory shocks as pure measurement error so they don't actually show up in the household problem.

[3] I did this to spare you from computing the stationary distribution over assets. If you had done that you could have drawn the initial distribution from that stationary distribution.